of publications shows a very strong positive effect ($\beta = 10.843$; SE = 0.238; z = 45.601; $p < 0.001$), and a positive and significant quadratic component ($\beta = 3.762$; SE = 0.119; z = 31.585; $p < 0.001$) indicates this relationship is nonlinear and accelerating. This means that as quantitative productivity increases, its marginal impact on the chances of entering the publishing elite becomes stronger.

Journal quality, measured by the average Scopus percentile rank and centered, has a positive and significant effect on the log-likelihood ratio of belonging to the top performer class ($\beta = 0.064$; SE = 0.005; z = 12.786; $p < 0.001$), which means that publishing in more prestigious journals increases the likelihood of entering the publishing elite even with an average production scale. However, the positive and very strong interaction between the logarithm of the number of publications and the average percentile of journals ($\beta = 0.185$; SE = 0.004; z = 42.086; $p < 0.001$) plays a key role. Since both variables have been centered, this parameter should be interpreted as an increase in the impact of quantitative productivity on the chances of achieving top performer status as the quality of publication channels rises above the average level.

This result clearly indicates that the quantity and quality of publications operate in a complementary rather than a substitutive manner. A high number of publications significantly increases the chances of belonging to the publishing elite, especially when these publications are located in high-prestige journals. On the other hand, intensifying production in low-prestige outlets brings relatively weaker selection benefits, and the high quality of individual publications alone, with a low scale of production, also does not maximize the probability of entering the class of top performers. The model thus empirically confirms that selection into the publishing elite is based on the cumulative interaction of quantity and quality.

Among the other control variables, the positive and significant effect of overall collaboration ($\beta = 0.517$; SE = 0.159; z = 3.254; $p \approx 0.001$) indicates that scientists with a higher-than-average level of co-authorship are more likely to achieve top performer status. International collaboration does not reach statistical significance ($\beta = 0.087$; SE = 0.077; z = 1.123; $p = 0.261$), which suggests that its independent effect disappears when the scale and quality of production are taken into account. The lack of a significant gender effect ($\beta = 0.012$; SE = 0.046; z = 0.266; $p = 0.791$) indicates that, after controlling for other factors, there are no gender differences in the likelihood of belonging to the publishing elite. Academic age has a positive and significant, albeit moderate, effect ($\beta = 0.010$; SE = 0.002; z = 4.373; $p < 0.001$), while the negative and significant effect of institution type ($\beta = -0.137$; SE = 0.047; z = −2.896; $p = 0.004$) indicates persistent structural differences between institutional contexts. The average team size shows a weak, negative, and marginally significant effect ($\beta = -0.008$; SE = 0.005; z = −1.789; $p = 0.074$), while the median journal prestige has a positive and significant effect ($\beta = 0.017$; SE = 0.003; z = 5.665; $p < 0.001$), suggesting that consistently high-quality publication channels bring additional selection benefits regardless of the average quality level.

**Table 7.** Model quality statistics

| Statistic | Value |
|---|---|
| n_obs | 264283 |
| n_authors | 144314 |
| logLik | -7755.794 |
| AIC | 15573.589 |
| BIC | 15898.617 |
| is_singular | FALSE |
| RE_var_AuthID | 0.029 |
| RE_sd_AuthID | 0.171 |

| | |
|---|---|
| R2_marginal | 0.978 |
| R2_conditional | 0.978 |
| ICC_AuthID | 0.009 |
| ICC_total | 0.009 |

The model was estimated on a large sample comprising 264,283 panel observations, corresponding to 144,314 unique scientists (Table 7). This number is smaller than in the original dataset because the analysis was performed on data aggregated to the level of author × period × discipline rather than on raw publication data. In addition, only observations with complete information in all explanatory variables were used for the estimation. Cases with missing data or nonfinite values resulting from transformations (in particular logarithmic and qualitative variable constructions) were automatically excluded at the stage of constructing the database used in the estimates, which is a standard procedure in mixed model estimation. The ratio of the number of observations to the number of scientists indicates a short panel structure in which the average scientist appears in more than one but still a small number of time-discipline cells. The number of scientists publishing for a long time – over all time periods – is extremely low, and our data show increases of the scientist population distributed over time periods.

The model fit was assessed based on the log-likelihood function and information criteria. The logLik value at the optimum is −7,755.79, while the AIC and BIC criteria reach 15,573.59 and 15,898.62, respectively. These measures are used for relative comparisons between alternative specifications estimated on the same sample and indicate a stable model fit given the number of parameters. It is also important that the model does not exhibit estimation singularity (is_singular = FALSE), as this means the random structure is correctly identified and the variance of the random effect has not been reduced to zero.

At the same time, the low ICC coefficient does not mean that the scientist's random effect is unnecessary. It indicates that, after taking into account the strong structural effects of period and discipline, the remaining individual heterogeneity is secondary, although still important for the correct modeling of intra-scientist relationships.

The random effect at the scientist level is characterized by a variance of 0.029, which corresponds to a standard deviation of 0.171 on the linear predictor scale. This translates into a very low intraclass correlation of approximately 0.009. In other words, after taking into account fixed effects, less than 1% of the total latent variance in the propensity to achieve top performer status can be attributed to permanent, unobservable differences between authors. At the same time, the positive value of this variance and the absence of singularities indicate that the random effect plays a corrective role, ordering intra-author relationships, but is not the main source of variation in the results.

The coefficients of determination were calculated according to the definition of Nakagawa and Schielzeth (2012), i.e., on the latent scale of the logistic model. The marginal coefficient of determination value of 0.978 indicates that almost all of the variance of the linear predictor is explained by fixed effects. A very similar value of 0.978 means that taking into account the random author effect increases the explained variance only minimally. Such a high $R^2$ value should not be interpreted in terms of the classic predictive fit known from linear regression but as a measure of the extent to which the identified structural factors determine the latent "propensity" to belong to the top performer class.

The high value is a direct consequence of the specificity of the phenomenon under study and the adopted construction of the dependent variable. Top performer status is relative and selective. Its definition is closely linked to the distributions of productivity and publication quality within periods and disciplines. At the same time, the model contains very strong structural effects of period and discipline and a clearly nonlinear component of quantitative productivity coupled with the quality of journals. As a result, a significant part of the latent predictor's variance is systematic and captured by fixed effects. Random differences between scientists play a secondary role.

Overall, the diagnostic properties of the model indicate its very good identifiability, estimation stability, and high ability to describe the mechanisms of selection to the publishing elite. The model does not exhibit a numerical problem, and the random structure is correctly estimated. The fit measures confirm that the key sources of variation in the probability of achieving top performer status are structural in nature and are adequately captured by the proposed specification. The diagnostic properties of the model confirm that the dominant source of variation in the probability of entering the publishing elite comprises structural factors – period and discipline – which provide an interpretative context for the strong, nonlinear effects of production quantity and quality.

Potential collinearity (Table 8) between explanatory variables was assessed using generalized variance inflation factors (GVIF), whereby, in accordance with the recommendations of Fox and Monette (1992), values adjusted for degrees of freedom were interpreted, that is, $\mathrm{GVIF}^{1/(2df)}$. This approach allows for comparison of the level of collinearity between predictors with different degrees of freedom, including multicategorical variables and interactions.

**Table 8.** Collinearity statistics

| term | GVIF | Df | GVIF_adj |
|---|---|---|---|
| Period | 2.689 | 4 | 1.132 |
| ln_pubs_c | 15.139 | 1 | 3.891 |
| ln_pubs2_c | 13.375 | 1 | 3.657 |
| mean_journal_pct_c | 11.852 | 1 | 3.443 |
| coop_c | 1.233 | 1 | 1.111 |
| coopint_c | 1.266 | 1 | 1.125 |
| Male | 1.049 | 1 | 1.024 |
| acage_c | 1.066 | 1 | 1.033 |
| inst_type | 1.014 | 1 | 1.007 |
| ATS_c | 1.249 | 1 | 1.118 |
| prestmed_c | 6.206 | 1 | 2.491 |
| ln_pubs_c:mean_journal_pct_c | 8.355 | 1 | 2.891 |

The adjusted GVIF values for most variables remain very low and close to unity. This applies to period effects ($GVIF_{adj}$ = 1.132 at df = 4) as well as to control variables describing collaboration, gender, academic age, type of institution, and average team size ($GVIF_{adj}$ = in the range of 1.007–1.125). These results clearly indicate the absence of significant collinearity in this part of the specification.

Inflated but still acceptable GVIF_adj values are observed for variables describing the scale and quality of publication output and their interaction. The logarithm of the number of publications reaches $GVIF_{adj} = 3.891$, its square component 3.657, the median percentile of journals 3.443, and the interaction of quantity and quality 2.891. A moderately increased value for median journal prestige (GVIF_adj = 2.491) reflects a partial co-occurrence of quality measures, it but remains well below conservative warning thresholds. Generally, these results indicate that the centering and construction of nonlinear components effectively reduced collinearity. The parameter estimates are stable and interpretable.

**Table 9.** Overdispersion statistics

| Statistics | Value |
|---|---|
| chi2 | 32124.81 |
| df_resid | 264252 |
| ratio | 0.122 |
| p_value | <0.001 |

The estimated logistic model was checked for excessive dispersion by comparing Pearson's chi-square statistic ($\chi^2$) with the number of residual degrees of freedom. In the analyzed case, the value $\chi^2 = 32{,}124.81$ at 264,252 degrees of freedom translates into a quotient of $\frac{\chi^2}{df} = 0.122$. Such a low value clearly indicates that the variability of the data does not exceed the level implied by the model assumptions (Table 9).

What is more, this result suggests the occurrence of so-called underdispersion, i.e., a situation in which the observed variability is lower than expected. In the context of a very large sample, a binary dependent variable, and strong structural effects included in the model (period and discipline), this result is intuitive and indicates a strong structural fit rather than a specification problem. The formal test does not provide grounds for rejecting the null hypothesis of variance consistency with the model assumptions ($p = 1$). This means that the estimated standard errors are not burdened with excessive residual volatility and statistical inference remains reliable. Consequently, there is no need for additional corrections or alternative dispersion specifications.

The distribution of Pearson residuals indicates a very good fit of the model at the observational level (Table 10). Although the extreme residual values reach −20.919 and 57.693, the central part of the distribution is strongly concentrated around zero (Q1 = −0.001, median = 0.000, Q3 = 0.000), and the mean residual is −0.003 with a standard deviation of 0.349. This means that for the vast majority of observations, the differences between the observed and predicted values are minimal, and a few outliers are incidental and do not indicate a systematic misfit of the model.

**Table 10**. Pearson residuals distribution

| Statistic | Value |
|---|---|
| pearson_min | -20.919 |
| pearson_q25 | -0.001 |
| pearson_median | 0.000 |
| pearson_q75 | 0.000 |
| pearson_max | 57.693 |
| pearson_mean | -0.003 |
| pearson_sd | 0.349 |

The classification ability of the model is very high, as indicated by an AUC value of 0.9985. This reflects a near-complete separation between observations belonging to the top performer class and those outside it in the predictor space. Such a result is consistent with the presence of strong structural effects of period and discipline, as well as with a pronounced, nonlinear, and complementary interaction between the quantity and quality of publication output.

Given the relative and selective construction of the dependent variable-defined as membership in the upper segment of the productivity distribution within period-discipline cells – a very high AUC value should not be interpreted as a sign of overfitting or diagnostic deficiency. Instead, it indicates that the model accurately captures the structural mechanisms governing entry into the publishing elite under the specified institutional, temporal, and disciplinary conditions.

As robustness checks (see Electronic Supplementary Material), we estimated a set of models alternative to the main specification (a logit model with period × discipline effects and quantity–quality controls) to assess whether the conclusions are driven by assumptions about the dependence structure in the panel and the functional form of productivity. First, alongside the subject-specific approach, we estimated a population-averaged logit using a generalized estimating equation approach (GEE) with clustering at the author level, which yields inferences that are robust to the random-effects specification and relies on the classical generalized estimating equations framework for correlated data (Liang & Zeger, 1986; Zeger et al., 1988). Second, to verify that the observed nonlinearities are not artifacts of imposing a polynomial in the log number of publications, we tested a more flexible functional form (e.g., splines/smooth terms), in line with standard practice in modeling nonlinear relationships in regression settings (Harrell, 2015; Hastie et al., 2009; Wood, 2017).

In addition, in a subset of analyses, we reestimated the models using alternative productivity definitions (e.g., variants based on fractional counting and different prestige weighting) to ensure that the results are not specific to a single metric. Across all these variants, the empirical pattern remained qualitatively stable: the core quantity × quality mechanism (a positive interaction effect) retained its sign and statistical significance, and the main conclusions regarding the differences between top performers vs. others did not change such that interpretation would be affected; differences concerned primarily the level of the effects (marginal, population-averaged interpretation in GEE vs. conditional interpretation in mixed models), as is expected when comparing subject-specific and population-averaged approaches (Zeger et al., 1988).

# 5. Discussion and conclusions

The aim in our analyses was to examine how research productivity and journal prestige jointly shape access to the publication elite (understood as membership in the top performer class). The analyses were based on a large bibliometric dataset covering articles indexed in the Scopus database over 30 years (1992–2021).

The unit of analysis was the scientist publishing in a specific six-year period and a specific discipline (author × period × discipline). The dataset included 433,546 unique research articles and 144,314 unique Polish authors. The analyses distinguished between a class of top performers, defined as the top decile of authors in terms of journal prestige-normalized productivity (the upper 10%), in a given discipline and period, and the class of remaining authors (90%) in a given discipline and period. The key measure of access to the publishing elite was the author's share of publications in journals belonging to the 90-99 Scopus journal percentile ranks. This measure was interpreted as a direct indicator of presence in the most prestigious channels of scientific communication. Author productivity was measured both by the number of publications and with the nonlinear journal prestige-normalized productivity index, which gives disproportionately greater weight to publications in top journals and reduces the weight of publications in bottom-level journals.

The most important result of the model is the strong and positive effect of the number of publications, the prestige of publication channels, and their interaction. Higher publishing intensity increases the chances of belonging to the publishing elite, especially when it is coupled with